Practical-1: What is Functional, Non-functional and Domain requirements? Explain the Functional, Non-functional and Domain requirements of the patan Multiple Campus library system in the required format.

State the scope of student enrollment management system of BIT. The computer based system can handle student enrollment process starting with application, exam and enrollment process. What information must be stored in the database during the process? State and explain Special terminology for library management domain requirements.

References:

Refer link: https://www.slideshare.net/RadikhaSharma/library-management-system-project-blackbook for functional and Non-functional requirements.

Sample of Functional requirements for Patient Appointment management System:

Functional Requirements

1. Manage Appointments
- 1.1. Patient makes new appointment.
- 1.2. Patient changes appointment.
- 1.3. Patient cancels appointment.

2. Produce Schedule
- 2.1. Office Manager checks daily schedule.
- 2.2. Office Manager prints daily schedule.

3. Record Doctor Availability
- 3.1. Doctor updates schedule

Nonfunctional Requirements

1. Operational Requirements
- 1.1. The system will operate in Windows environment.
- 1.2. The system should be able to connect to printers wirelessly.
- 1.3. The system should automatically back up at the end of each day.

2. Performance Requirements
- 2.1. The system will store a new appointment in 2 seconds or less.
- 2.2. The system will retrieve the daily appointment schedule in 2 seconds or less.

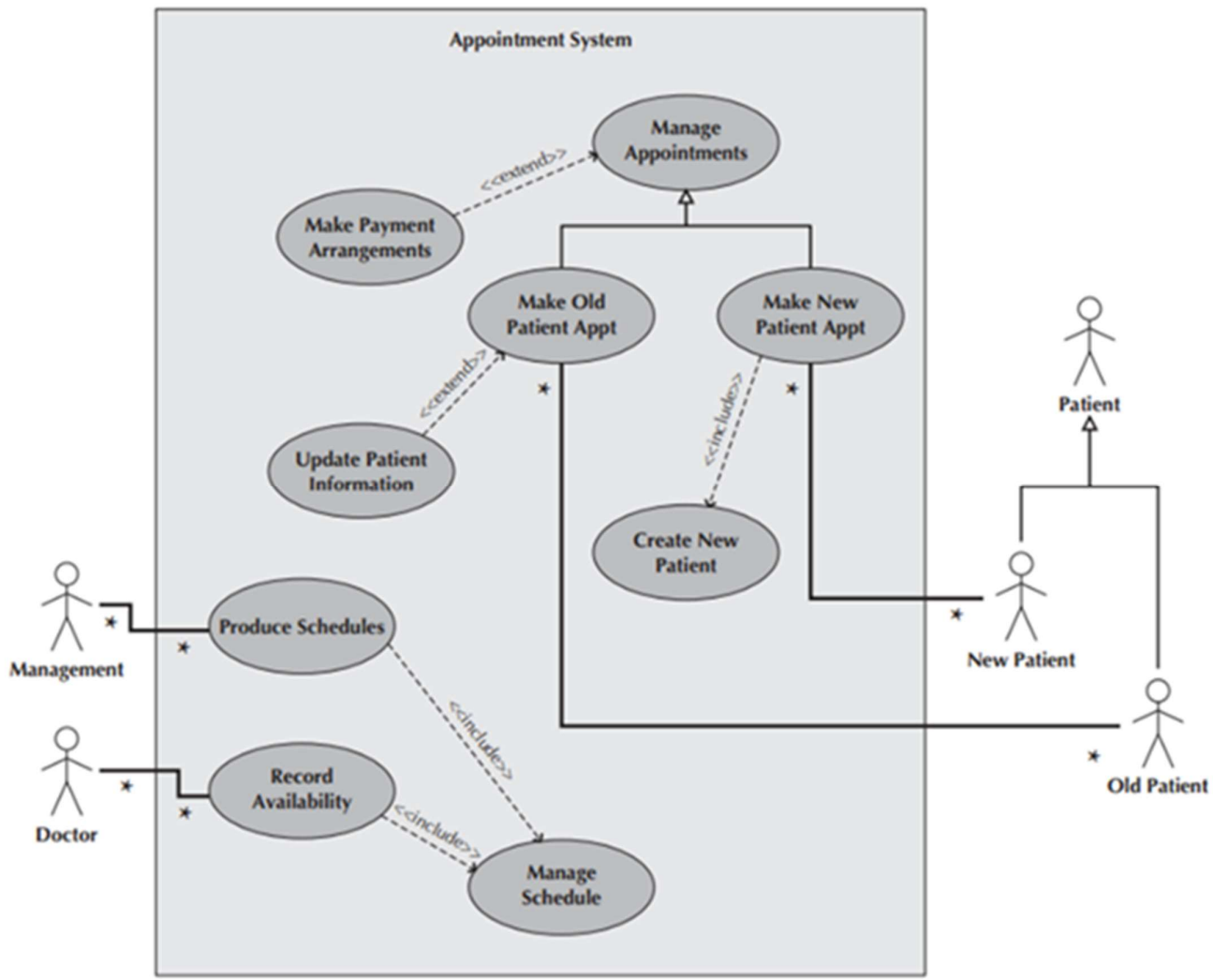
3. Security Requirements
- 3.1. Only doctors can set their availability.
- 3.2. Only a manager can produce a schedule.

4. Cultural and Political Requirements
- 4.1. No special cultural and political requirements are anticipated.

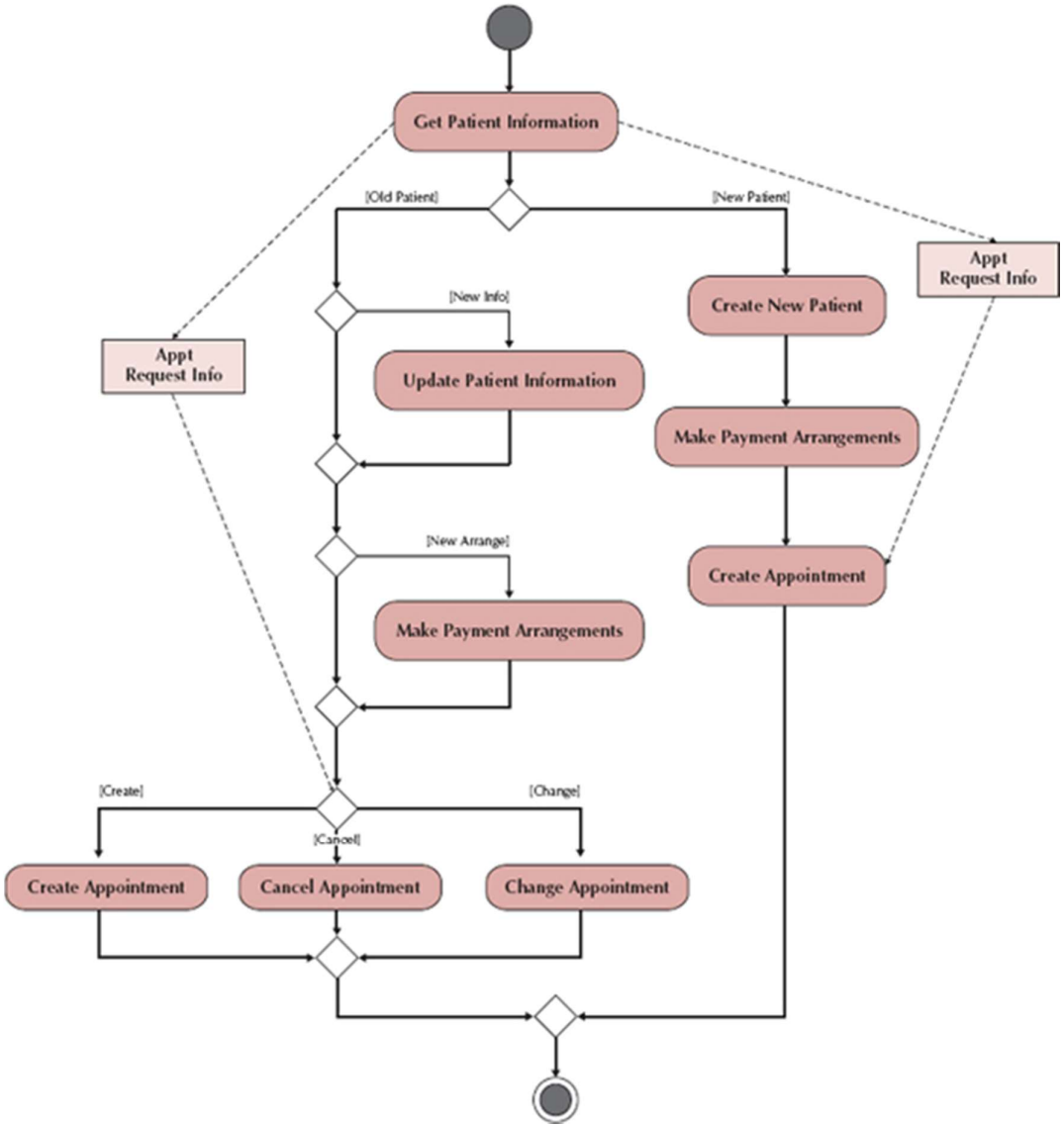
Practical-2: What is system modeling? Explain different perceptives of system modeling.

objective: Use Draw.io or any other CASE ools to draw the following diagrams and prepare PDf file to import in word document.

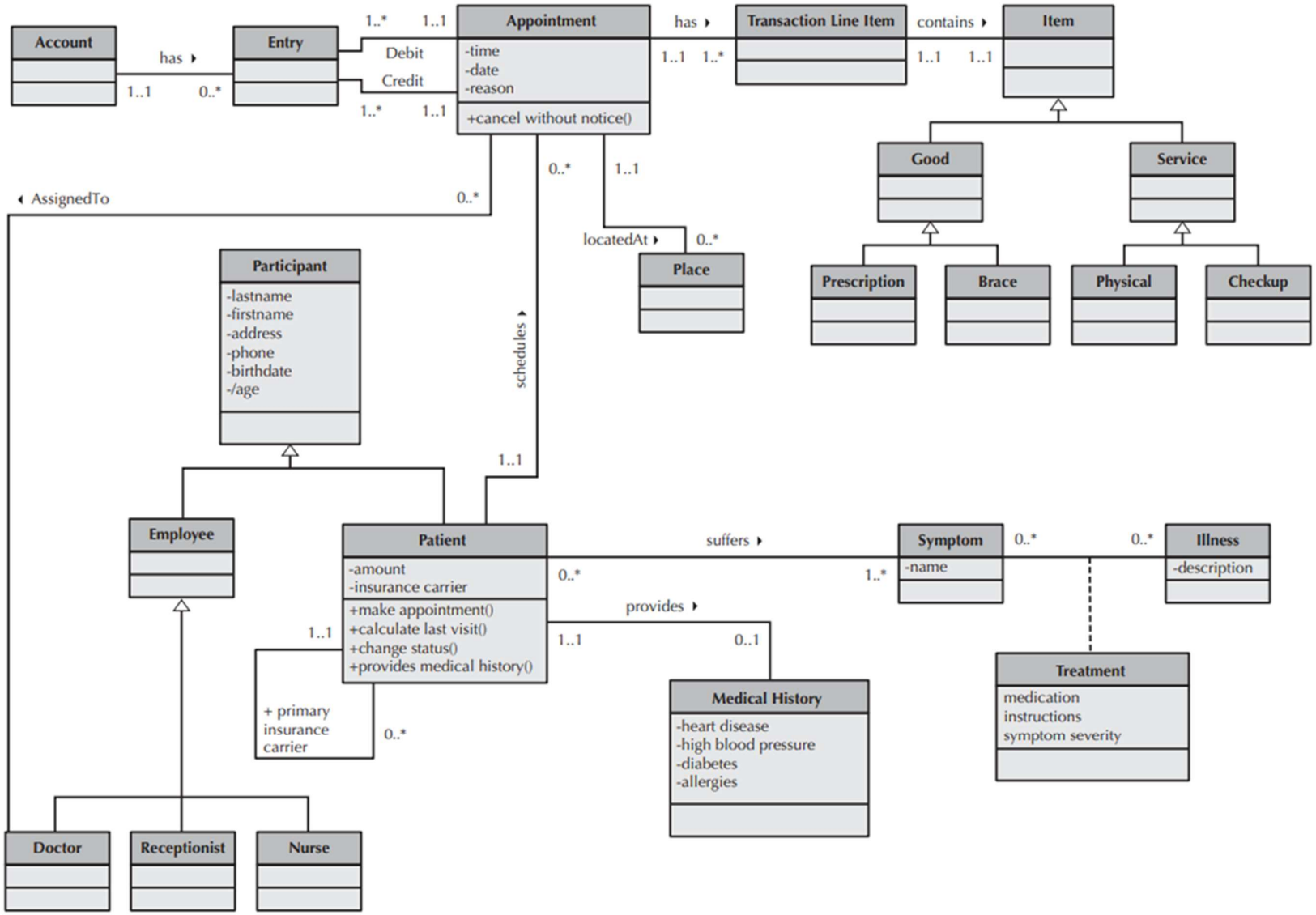
a) Use case diagram



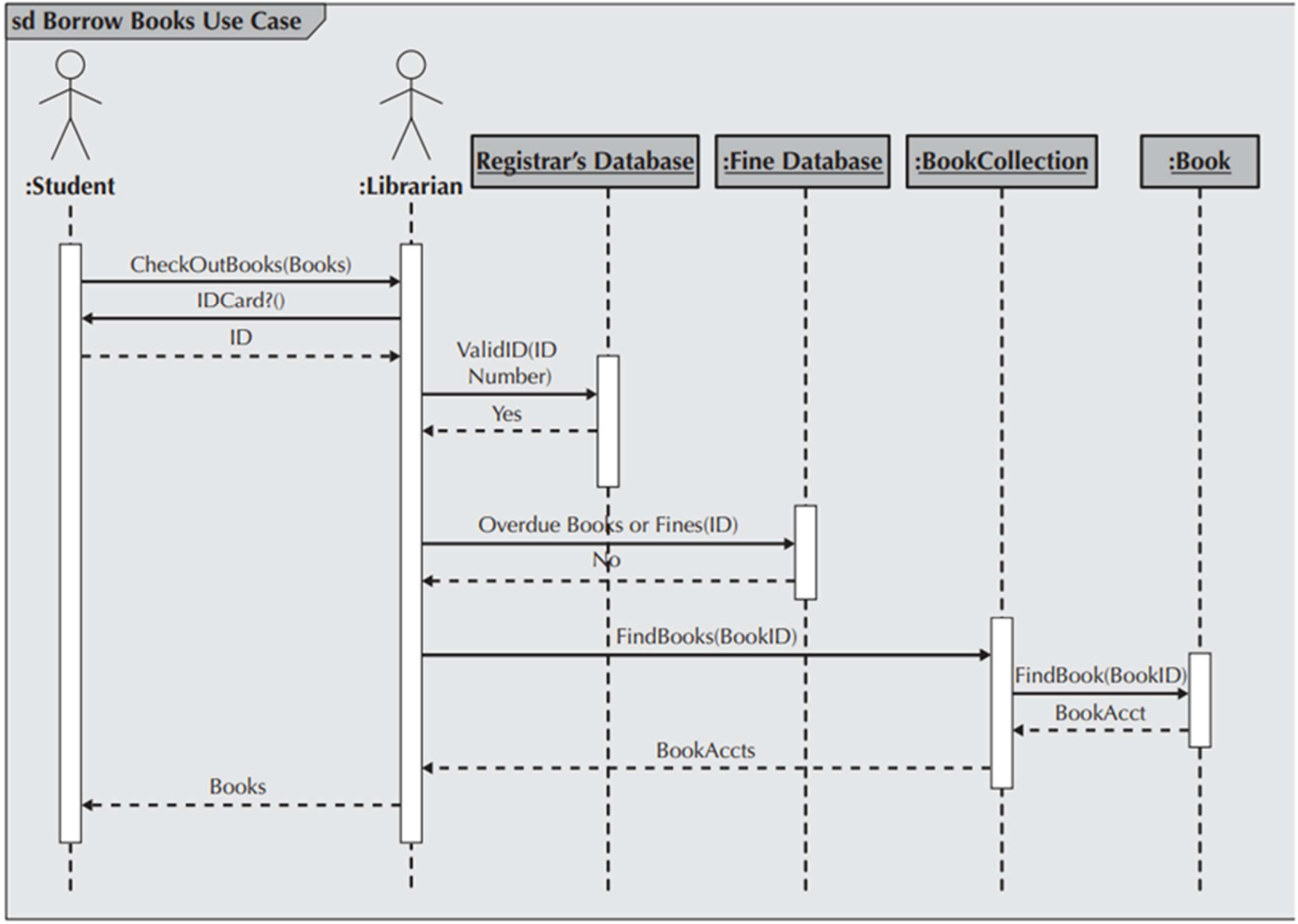
b) Activity Diagram



c) Class Diagram



d) Sequence diagram



e) Communication Diagram

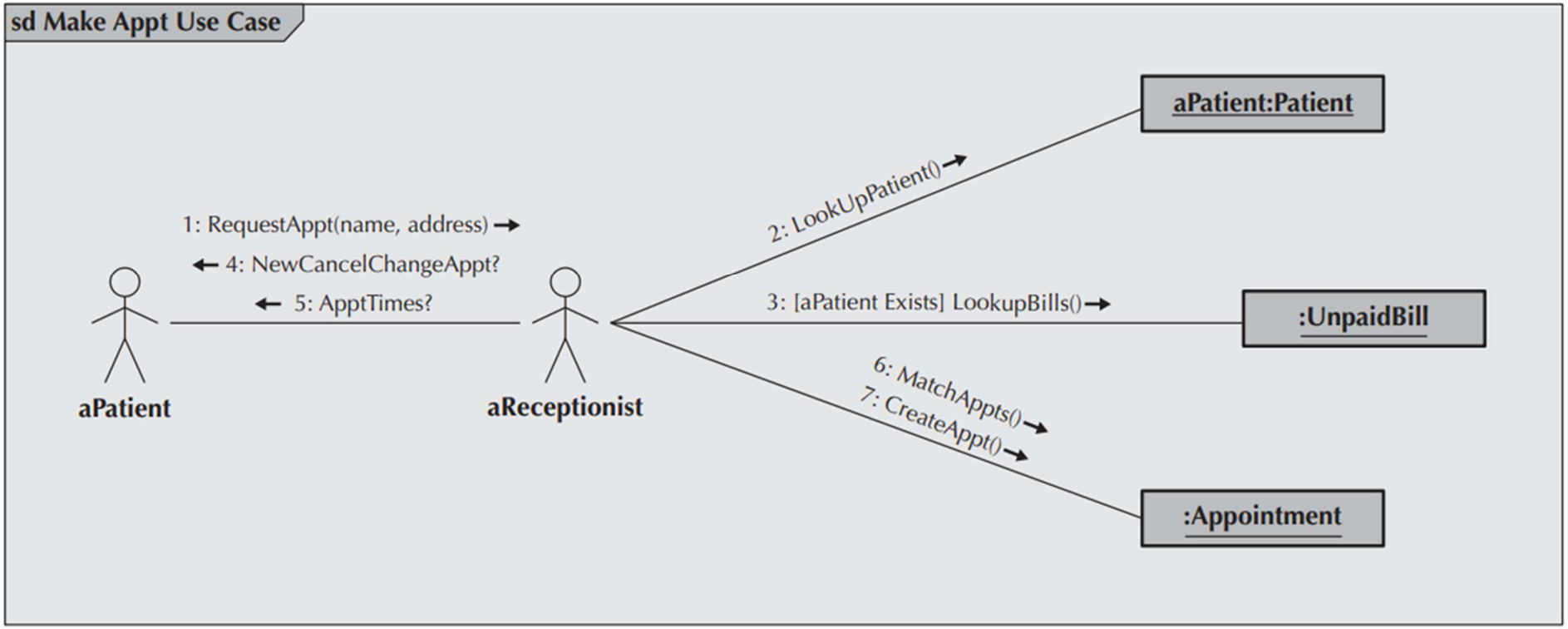


FIGURE 6-10 Sample Communication Diagram

Practical-3: Install **Git** in your laptop and set up a **GitHub** account. All students should be capable to do following in using the **GitHub.**

Refer link: How to Use Git and Git Workflows – a Practical Guide (freecodecamp.org)

- Install Git and set up a github account
- How to create a new repository in GitHub
- Cloning the repository
- Git branches
- How the check the status of a Git project
- How to make first commit
- How to push up first commit to GitHub
- How to add another commit in Git
- How to stage changes in Git
- How to view the Git difference
- How to collaborate with others in Git
- Feature branches in Git
- Git workflows for collaboration
- How to merge a branch in Git
- Pull request workflow
- How to bring local up to date
- How to retrieve remote data

Practical-4: Familiarization of JIRA. All students should be capable to use JIRA, a powerful tool for project management and issue tracking across diverse teams and projects.

Students should know the some key functionalities of JIRA as below:

- **Issue Tracking:** JIRA excels in managing and tracking various types of issues, tasks, and projects.
- **Customizable Workflows:** It allows teams to define and customize workflows to match their processes.
- **Dashboards and Reports:** Provides visual dashboards and reports to track team progress and project metrics.
- **Integration Capabilities:** Integrates seamlessly with various development tools like Git, Confluence, etc.
- **Agile Project Management:** Supports Agile methodologies such as Scrum and Kanban.

Practical-5: Familiarization of software project management (MS Project).